WHAT LOOKS LIKE A CAPABILITY LIMIT IN VISION–LANGUAGE MODELS IS A READOUT LIMIT

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ABSTRACT

Benchmarks for vision–language models offer their answer choices in some convention: a letter, a color name, a pixel coordinate. That convention is treated as neutral. We find it is not, and that the limits a benchmark reports can belong to the readout rather than to the model.

On 200 COCO photographs, Qwen3-VL-4B picks the correct one of nine locations for a named object 68.5% of the time when the locations are given in English and 20.0% when the same nine locations are given as pixel coordinates. Chance is 11.1%. The cost arises when the answer options are coordinates; giving the model a coordinate in the question instead costs 3.5 points and is not significant. The gap holds on a 4×4 grid, under 8-bit rather than 4-bit quantization, and in every slice by object size, boundary distance and category. It also decides which model wins. Two models that tie under English names differ by 39 points in one coordinate system and by 54 in the other, in opposite directions.

On the color task, three of the four open models that can do the task at all show the penalty; on photographs, two of three open models do, and so does Gemini, at 11.1 points on parseable answers ($p = 10^{-4}$). GPT-4o does not. To ask whether a model reads a coordinate at all, we attach the wrong name to each one and record which the model follows. Color options written as hue angles are followed below chance; a normalized pixel convention is followed at four times chance. This tells apart conventions a model can use from ones it cannot, although it did not predict accuracy on two conventions we had not tried. Five models also name the same color wheel five different ways, so a fixed answer vocabulary is not neutral across models either.

Five times during this work we measured a capable model as incapable because our scorer and the model disagreed about what an answer looks like. We report each case. They are the phenomenon in miniature.

1 INTRODUCTION

Every benchmark for a vision–language model makes a choice it rarely discusses: how the answer is to be written. A question about color may be answered with a word or a hue angle; a question about position with a phrase, a grid reference, or a pair of pixel coordinates. The benchmark author picks one, the score is computed against it, and the number is reported as a property of the model. The convention does not appear in the number.

We show that it should. On 200 photographs from COCO, Qwen3-VL-4B is asked where a named object is and offered the nine cells of a 3×3 grid. When those nine cells are described in English it answers correctly 68.5% of the time. When the same nine cells are described as pixel coordinates, in a prompt that also states the image size and the axis conventions, it answers correctly 20.0% of the time. Chance is 11.1%. Nothing about the image, the object, the question, or the set of alternatives has changed. The convention alone accounts for 48.5 points.

The loss is not symmetric. If the model is instead handed a pixel coordinate and asked which object sits there, the same convention costs 3.5 points and is not significant. A coordinate on the answer side is what costs the model; on the question side it costs little. This matters because spatial

benchmarks put coordinates on the answer side, and a model that scores poorly on them may know perfectly well where the object is.

The convention also decides comparisons between models. Qwen2.5-VL-7B and Qwen3-VL-4B are statistically tied under English cell names, 68.0% against 68.5%. Under pixel coordinates the older model leads by 39 points; under a normalized coordinate convention the newer one leads by 54. Both differences are highly significant and they point in opposite directions. A leaderboard built on either convention would rank these models confidently and, depending on the convention, oppositely.

We then ask why a convention fails. Attaching an English name to each pixel coordinate appears to repair the readout. Attaching the *wrong* name shows what actually happened: the model follows the name and abandons the coordinate, and for hue angles it follows the coordinate less often than chance. The coordinate was barely read. A normalized pixel convention, by contrast, is followed four times more often than chance even against a contradicting name. We call the share of responses that follow the coordinate, relative to chance, the convention's *readership*. It separates conventions a model can use from ones it cannot. When we pre-registered a test of whether it also predicts accuracy on two new conventions, it did not, and we report that.

Finally, the problem is not specific to coordinates. Asked to name the colors of a calibrated wheel, five models produce five different stable vocabularies: only one has a red, only one a teal, and one admits just four names. A fixed set of color words, the ordinary format of a color question, is therefore not a neutral instrument across models either.

Contributions.

- A convention swap that holds the image, the question, the alternatives, their order and the output format fixed and varies only how the alternatives are described, applied to color and to location, on synthetic stimuli and on real photographs (§4, §5).
- Evidence that the convention changes not just scores but rankings, with two significant rank inversions among three open models on identical images (§7).
- A mispaired-name test that measures whether a convention is read at all, together with an honest account of what it does and does not predict (§6).
- Replication on frontier models: the gap shrinks with capability and survives it in Gemini; it is absent in GPT-4o (§7).
- Per-model vocabulary calibration, and the finding that five models partition one color wheel five different ways (§3.1).
- An audit of our own scoring in which we five times measured a capable model as incapable, reported as evidence for how easily this error is made (§8).

We do not claim that all benchmarks are wrong or that every coordinate system is unreadable, and where we attribute a convention to a model's training we cite the model's own report. We claim that the answer convention is a variable, that it is often large, and that it is currently invisible.

2 Related work

The answer interface. The closest prior work is Wang et al. (2026b), which names the problem in nearly our terms: spatial benchmarks demand coordinates, options or text, and call the resulting gap an answer-interface mismatch. Their remedy is to change the output modality, letting image-generation models draw their answers and parsing the drawings back into the original metrics. We keep the text interface fixed and change only how the alternatives are described, which lets us compare conventions on identical items, and we add the mispaired-name test to ask whether a convention is read at all. Neither the paired comparison nor the mispaired-name test appears in that work.

Benchmarks that are too easy. Lan et al. (2026) and Yuan et al. (2026) show that many benchmark items can be answered without the image, from leakage in the question text or from world knowledge. Our finding runs the other way: a benchmark can be too hard for a reason that has nothing to do with the model's vision. The two distortions are not exclusive. A single reported number

can be inflated by leakage and deflated by convention at the same time, and neither is visible in the number. Singla et al. (2026) varies the phrasing of the question to measure reliance on textual priors; we hold the question fixed and vary the description of the answers.

Selection bias in multiple choice. A well-developed line of work shows that language models and vision–language models prefer particular option positions and particular option tokens, and mitigates this by permuting order (Liu et al., 2024), debiasing the prior over labels (Zheng et al., 2024), or separating positional from token effects (Atabuzzaman et al., 2025). Every method in this line holds the option *descriptions* fixed. In our design that corresponds to the single rung in which a color word is replaced by an integer label pointing at the same word, and that rung costs one point. The thirty-point drop occurs one step earlier, when the description itself changes, and selection bias methods do not reach it. Cappelletti et al. (2025) show that first-token scoring can be made reliable by prefilling the answer prefix; our audit reaches the same conclusion from the failure side, by refusing to score any alphabet the model does not place mass on.

Grounding models. Systems such as Rasheed et al. (2024) and benchmarks such as Cheng et al. (2025) treat coordinate output as a capability to build and to measure. We do not disagree. We measure coordinate output as an instrument that can misreport a capability the model has, and our framework predicts that a model trained to emit coordinates will show a smaller convention gap. A separate line treats the textual coordinate readout itself as the defect and replaces or retrains it: explicit position-to-coordinate tokens (Wang et al., 2025), attention-based pointing with no coordinate tokens at all (Wu et al., 2025), parallel box decoding (Wang et al., 2026a), and distillation against geometric rather than token loss (Zhu et al., 2026). Those are remedies on the model side; ours is the measurement they presuppose, of how much a fixed model's score depends on the readout it is handed.

Color stimuli. The wheel is the CIELAB circle at fixed lightness standard in human visual working memory research (Zhang & Luck, 2008; Bays & Husain, 2008), used for its perceptual uniformity; the paper makes no comparison to people.

3 METHOD

Every experiment holds the image, the question, the candidate set, the candidate order and the required output format fixed, and varies only how the candidates are *described*. We call that manipulation a *convention swap*. Three further components make it interpretable: a response vocabulary calibrated per model, a capability gate, and a held-out split declared before the data existed.

3.1 CALIBRATING THE RESPONSE VOCABULARY

A convention swap needs a control condition the model can certainly express. We do not assume one. For each model we render one swatch per wheel angle on a 5° grid, three times per angle under different square sizes and placements, and ask it to name the color in free report. An angle is retained only if all three renders elicit the same word. Anchors are the centers of contiguous unanimously named runs, separated by at least 40°.

Calibration yield is an outcome, not a setting. The five models tested here do not agree. Qwen3-VL-4B's wheel contains no red: the swatch a coordinate table calls red renders `(238,146,146)`, and the model calls it pink, correctly. SmolVLM-2B's wheel does have a red. Qwen2.5-VL-7B has a stable teal where Qwen3-VL has none. A fixed textual answer set therefore need not carry equivalent perceptual grounding across models (Table 5, Appendix C).

We also resolve the *surface* of each candidate token per model rather than assuming it. Candidate ids are taken from whichever surface variant (bare, leading space, capitalized, and their combinations) places the most probability mass on the candidate set under probes with known answers. A model whose candidate mass never clears 0.5 has no restricted first-token readout for that alphabet and is refused rather than reported. Section 8 explains why this guard exists.

3.2 THE CONVENTION SWAP

Within a trial the model chooses among K alternatives that denote the same K referents in every condition. Only their surface description changes: a color word versus a hue angle, an English spatial phrase versus a pixel pair. Output format is held fixed within a comparison, so the cost of emitting a digit instead of a word is measured separately and does not contribute; it is 1 point (Table 1).

3.3 VISIBLE AND CACHED CONDITIONS, AND THE CUE CROSSOVER

For the color experiments each trial is run two ways. In the *visible* condition the array is in the prompt with the question. In the *cached* condition the array is encoded once into the model's key–value cache with an acknowledgement turn, and the question is a text-only continuation with no image present; the model answers from what it retained. Both are scored the same way.

To check that a cached answer depends on the remembered binding at the cued location, and not on the array as a whole, we also run a *cue crossover*. Two arrays differ only by swapping the colors at two locations. Asking about the first location should shift the score contrast one way, asking about the second should shift it the other way, and asking about a third, untouched location should not shift it at all. The reported *interaction* is the difference between the first two shifts, normalized by the spread of the candidate scores so that conditions with different score ranges can be compared. A location-blind account predicts no sign reversal.

3.4 THE MISPAIRED GLOSS, AND READERSHIP

Attaching a name to a coordinate, as in `1=(66,50) [top-left]`, raises accuracy, but accuracy alone cannot say whether the coordinate was used or merely accompanied. We therefore *derange* the gloss so that no option carries its true pairing. Every coordinate and every name still appears exactly once; only the pairing breaks, and the derangement is verified to have no fixed point, so the two resulting keys can never coincide. One response is then scored against both: **key_coord**, the option whose coordinate is correct, and **key_name**, the option whose glossed name is correct.

We define *readership* as

$$\rho \;=\; \frac{P(\text{response} = \text{key_coord} \mid \text{keys disagree})}{1/K}, \tag{1}$$

the share of responses following the coordinate, relative to chance. $\rho \approx 1$ means the coordinate carries no information the model uses; $\rho < 1$ means it is actively avoided, since the correct coordinate now sits beside the wrong name.

3.5 CAPABILITY GATE

A convention gap is a difference between two readouts of the same competence. Where there is no competence, a null contrast is uninformative rather than disconfirming. We therefore require the control condition to exceed chance by 20 points *on the readout the contrast is computed with*. Gating on a different readout than the contrast uses admits models whose contrast is measured by an instrument with no signal in its own baseline; we made exactly that error and corrected it.

3.6 HELD-OUT SPLIT

Because the vocabulary is chosen by asking the model, evaluating at the same angles cannot separate "the model has this representation" from "we selected the colors it likes". The held-out rule was committed before the generator was written: integer angles 10–20° from each anchor, off the calibration grid, and at least 20° from every other anchor. All hypotheses, endpoints, exclusions and failure conditions in this paper were registered in a versioned analysis plan before the corresponding code existed; two of our own predictions were refuted by it and are reported as such (Sections 4 and 6).

Table 1: Color, Qwen3-VL-4B, 173 held-out trials. Only the description of the six alternatives changes; stimulus, candidate set, candidate order and output format are fixed. Interaction is the spread-normalized cue-crossover statistic (§3.3); chance is 16.7%.

options described as	visible	cached	interaction	% of word
color word	84.4	85.0	9.27	100
integer label, word listed	83.8	83.5	8.99	97
hue angle with the word attached	77.5	80.1	8.22	89
hue angle, legend stated once	53.8	63.2	6.34	68
hue angle alone	21.8	22.3	1.08	12

4 COLOR: THE CONVENTION GRADIENT

We first establish the effect where every factor can be controlled. Stimuli are four colored squares on a gray field, drawn from a vocabulary calibrated per model (§3.1); the model is asked which color occupied a cued location. All figures below are on held-out angles, never used to select the vocabulary, sampled by a rule committed before the generator was written.

Three facts in Table 1 matter. Changing the *output* from a color word to an integer costs 1 point, so what follows is not about emitting digits. Changing the *description* of the same six alternatives from words to hue angles costs 30.1 points visible (175 label-only correct against 19 numeric-only, exact McNemar $p = 8.7 \times 10^{-33}$, $n = 519$). And a hue angle with no grounding at all sits at 21.8% against 16.7% chance: it is not a degraded readout, it is not a readout.

Accuracy and the cue-crossover interaction move together across all five rows. The crossover swaps two remembered colors between two cued locations and predicts a sign reversal that a location-blind account cannot produce; a third, untouched location is null at every row (-0.55 to $+0.34$). That the evidence measure tracks accuracy rules out an account in which only response formatting changes. The cached column shows the same gradient with no image in context: the convention, not the availability of the pixels, sets the score. On the calibration angles themselves the word conditions score near 99%; the drop to 84% on held-out angles is the price of the split, and the reason it exists. No benchmark scores color in CIELAB hue angles; the hue rows are the limiting case of a convention with no grounding at all.

Grounding is not what the gloss supplies. Attaching the word to the angle appears to repair the readout, and does not. Deranging the gloss so that no option carries its true pairing separates the two keys: the model follows the *name* on 75.1% of cells and the *angle* on 5.1%, against 16.7% chance ($p = 7.7 \times 10^{-21}$ below chance, $n = 692$). The correct angle is not ignored but actively avoided, because it now sits beside the wrong word. The apparent recovery is substitution.

5 SPACE, AND REAL PHOTOGRAPHS

If the deficit were about color it would stop here. We repeat the manipulation on location, first on synthetic arrays and then on COCO val2017 photographs (Lin et al., 2014). On the synthetic arrays, four colored squares in nine cells, the locate ordering is cell name 70.8%, normalized 58.3%, pixel 42.5%, and the same with the array cached (Table 8, Appendix C). On COCO, one object per image, its position asked for as a choice among the nine cells of a 3×3 grid; only the description of those nine cells varies. Admission criteria were fixed before the data was touched: a unique instance of the category, segmentation area under 25% of the image, and a center at least 8% of the image inside its cell's border.

A coordinate on the answer side is costly; on the question side it is almost free. Table 2 is the paper's central result. Asking where an object is, with the nine candidates written in pixels, scores 20.0% against 68.5% in English, a drop of 48.5 points (95% bootstrap interval $[-56.5, -40.0]$, $p = 1.2 \times 10^{-21}$) on a task whose chance rate is 11.1%. Handing the model the same pixel coordinate as *context* and asking what is there costs 3.5 points and is not significant ($p = 0.065$). The asymmetry replicates in InternVL3-2B (-32.5, $p = 4.2 \times 10^{-13}$ on the answer side; -8.0 on the question

Table 2: COCO val2017, 200 photographs. LOCATE asks for the position of a named object (9 options, chance 11.1%); IDENTIFY supplies the position and asks which object is there (6 options, chance 16.7%). Δ is against English cell names, exact McNemar; *** $p < 10^{-3}$. The deranged row is scored on the coordinate key.

	LOCATE		IDENTIFY	
options described as	acc.	Δ	acc.	Δ
English cell name	68.5	—	96.0	—
normalized 0–999	59.5	-9.0	97.0	$+1.0$
absolute pixels	20.0	-48.5^{***}	92.5	-3.5
pixels, name attached	39.0	-29.5^{***}	95.5	-0.5
pixels, name deranged	16.0	-52.5^{***}	93.5	-2.5

Table 3: Which key does the model follow when coordinate and name disagree? Readership is the share following the coordinate, relative to chance. Qwen3-VL-4B throughout; n is scored cells.

convention	follows coord.	follows name	neither	readership	n
CIELAB hue, name inline	5.1%	75.1%	19.8%	$0.30\times$	692
CIELAB hue, legend	8.2%	61.7%	30.1%	$0.49\times$	692
pixels, synthetic grid	21.7%	31.7%	46.7%	$1.95\times$	120
pixels, COCO	16.0%	25.0%	59.0%	$1.44\times$	200
normalized 0–999, COCO	45.0%	14.0%	41.0%	$4.05\times$	200

side). On a 4×4 grid with sixteen cells and 6.25% chance, cell names score 26.0% and pixels 7.5% ($p = 3.0 \times 10^{-6}$): at finer granularity pixels fall to chance while names stay four times above it. The gap holds within every tercile of object size and of distance from a cell boundary, and in eleven of twelve COCO categories with at least five images (Appendix C). The pixel prompt states the original image size; the vision encoders rescale by a few percent at most, less than the 8% margin between a cell center and its border, so the frame cannot move an answer across a cell.

Prompt length is ruled out from inside the design. In the synthetic spatial task the normalized-coordinate prompt is *longer* than the pixel prompt, 187 tokens against 183, and scores 15.8 points higher. No external control is required.

The readable convention is model-specific. The normalized 0–999 convention nearly rescues Qwen3-VL (59.5% against 20.0% for pixels) and barely helps InternVL3-2B (19.0% against 13.5%). Qwen3-VL's technical report describes a normalized $[0, 1000]$ grounding system (Qwen Team, 2025); the advantage of that convention does not transfer to a model trained differently. This mirrors what calibration already showed for color: models do not share an answer vocabulary, and they do not share a readable coordinate system either.

6 READERSHIP: IS THE CONVENTION READ AT ALL?

Accuracy under a convention conflates two things: whether the model can use the convention and whether it happens to reach the right answer by some other route. The mispaired-name test of §3.4 separates them. Each coordinate is glossed with a name that belongs to a different option, and one response is scored against both keys.

Hue angles are avoided, not ignored. With a mispaired name beside it, the correct hue angle is chosen on 5.1% of cells against a chance rate of 16.7% ($p = 7.7 \times 10^{-21}$ below chance). The model is not indifferent to the coordinate; it moves away from the slot carrying the correct angle because that slot now carries the wrong word. The same holds when the pairing is stated once in a legend rather than inline. What looked in Table 1 like a gloss repairing the readout was the gloss replacing it.

A familiar convention is read even against a contradiction. The normalized 0–999 convention is followed on 45.0% of mispaired trials against 11.1% chance, and the name on only 14.0%. The preference reverses completely between the two ends of the table. Nothing differs between those rows except the units, and the model's behavior with the units when they disagree with the words is the cleanest evidence we have that some conventions are read and others are not.

What readership does not do. Before running two further conventions we registered the prediction that accuracy would be monotone in readership and placed each new convention relative to the pixel and normalized anchors. A grid reference ($A1$ to $C3$) had readership $1.26\times$ and a percentage convention $1.30\times$, both below pixels, so both were predicted to score under 20.0%. They scored 31.0% and 23.0%, in the wrong order; the prediction failed on every part. The reason is visible in the table: readership mixes how usable a convention is with how hard a competing English name pulls, and the grid reference has the strongest name pull of any convention tested on photographs. We therefore report readership as a test of whether a convention is read, which it is, and not as a predictor of accuracy, which it is not, and we did not search for a statistic that fits the five points we have.

Readership for each model on COCO pixels is given in Appendix A.

7 Across models

Five vocabularies for one wheel. Each model was calibrated separately by the procedure of §3.1. Table 5 (Appendix C) shows what they returned. Green near $145°$ is the only anchor all five share. Qwen3-VL's wheel has no red; SmolVLM's does. Qwen2.5-VL stabilizes a teal that Qwen3-VL, one generation later in the same family, cannot. LLaVA-1.5 admits four stable names where the others admit six. A color question with a fixed set of answer words is asking different models different questions.

The color gap across models. On held-out angles with each model's own vocabulary, the word-to-numeric penalty is 30.1 points in Qwen3-VL-4B ($p = 8.7 \times 10^{-33}$), 23.9 in Qwen2.5-VL-7B ($p = 9.1 \times 10^{-4}$) and 19.7 in InternVL3-2B ($p = 1.6 \times 10^{-2}$). LLaVA-1.5-7B clears the capability gate and shows no penalty (-2.1, $p = 1.0$); it is the oldest model tested and the one with the fewest stable colors, which we report as description and not as explanation. SmolVLM-2B does not clear the gate and is uninformative. InternVL3 shares a Qwen language backbone, so it is a partial cross-family replication and we describe it as one.

Real photographs, five models. Figure 1 gives the locate task on the same 200 COCO images for three open models run locally and two frontier models run through their APIs (`gemini-flash-latest` and `gpt-4o` as served on 20 September 2026). All saw byte-identical prompts. LLaVA-1.5 and SmolVLM are absent because the harness refused to score them: no first-token digit surface carried probability mass (§8). The frontier models are scored by generate-and-parse because their APIs expose no usable first-token distribution; the local models are scored by first token and, where both are available, the two readouts agree on 97–100% of trials (Appendix B). Every headline p-value is below 10^{-3} except InternVL3-2B's color gap.

Quantization makes the gap larger, not smaller. Running Qwen3-VL-4B at 8-bit rather than 4-bit raises every condition and widens the gap from 48.5 to 52.0 points. The effect is not an artifact of low precision.

The gap survives at the frontier, and shrinks. Gemini scores 94.5% under English names and still loses 11.1 points to pixel coordinates on parseable answers. GPT-4o, which is weak at this task under every convention, shows no significant gap. The convention penalty is largest in small open models, smaller in a frontier model, and present in one of the two frontier models we could test.

Rankings reverse. Table 4 pairs the models on the same images. Qwen2.5-VL-7B and Qwen3-VL-4B are tied under English names. Under pixels the older model leads by 39 points; under the normalized convention the newer one leads by 54. InternVL3-2B trails Qwen2.5-VL under names and pixels and leads it under the normalized convention. Two of three pairs invert. We registered the

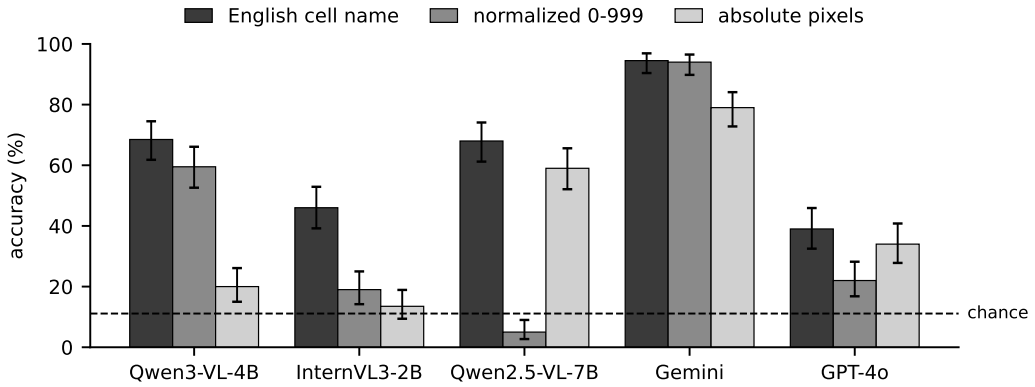


Figure 1: COCO LOCATE on the same 200 photographs; chance is 11.1% (dashed). Bars are accuracy with 95% Wilson intervals. The cell-name minus pixel gap, paired per image with a 95% bootstrap interval over images, is -48.5 $[-56.5, -40.0]$ for Qwen3-VL-4B, -32.5 $[-40.5, -24.5]$ for InternVL3-2B, -9.0 $[-18.5, 0.0]$ for Qwen2.5-VL-7B, -15.5 $[-21.5, -10.0]$ for Gemini and -5.0 $[-12.5, 2.5]$ for GPT-4o. Exact values, the 8-bit run and the parseable-only analysis are in Table 6, Appendix C.

Table 4: Paired differences on the same 200 photographs, exact McNemar. A sign change across rows within a pair is a rank inversion.

pair	convention	difference	p
Qwen2.5-VL-7B vs Qwen3-VL-4B	cell name	-0.5	1.0
	pixel	$+39.0$	8.6×10^{-14}
	normalized	-54.5	1.2×10^{-30}
InternVL3-2B vs Qwen2.5-VL-7B	cell name	-22.0	5.7×10^{-7}
	pixel	-45.5	8.5×10^{-19}
	normalized	$+14.0$	1.5×10^{-5}
InternVL3-2B vs Qwen3-VL-4B	cell name	-22.5	3.6×10^{-7}
	pixel	-6.5	0.07
	normalized	-40.5	1.9×10^{-22}

search for inversions before running the models and report every pair, inverting or not. The direction of the Qwen inversion follows the two models' documented training conventions: Qwen2.5-VL's report states that it represents boxes and points in coordinates based on the actual dimensions of the input image (Bai et al., 2025), and Qwen3-VL's describes a normalized $[0, 1000]$ system (Qwen Team, 2025). Each model leads in the convention its own report says it was trained in.

Qwen2.5-VL-7B's 5.0% under the normalized convention is below chance for the same reason: on 53.0% of trials it chooses the option whose 0–999 numbers, read as pixels, lie nearest the object (null 11.8%, $p = 7.8 \times 10^{-48}$), and that option is the true target only 2.5% of the time. Qwen3-VL-4B shows the mirror on pixels, weaker; InternVL3-2B shows neither. The test was post hoc (Appendix A).

8 A READOUT AUDIT OF OUR OWN EXPERIMENTS

Five times in this work we scored a model as performing at or near chance, looked at what it had actually produced, and found it answering correctly. In each case the scorer and the model disagreed about what an answer looks like. We report them because they are the paper's claim at the scale of a single line of code, and because each one was invisible from the score.

The wrong token surface. SmolVLM-2B scored 16–30% against 16.7% chance on every color condition, with candidate mass 0.00. It was emitting `Green`, leading space and capital, token 4910; the scorer read `green`, token 9298. Its top three first tokens on a green swatch were `Green`, `Red`, `Blue`, the correct one far ahead. The fix is to resolve each model's emitted surface by measurement, and to refuse to report when no surface carries mass.

No first-token readout exists. For SmolVLM and LLaVA-1.5 every integer label begins with the same shared leading-space token. A restricted first-token score over such labels is not noisy; it does not exist. The harness now refuses rather than emitting chance-level numbers, and those models are scored by generating and parsing.

Held-out stimuli in a discarded category. The held-out rule kept color angles 20° from every retained anchor but not from names the calibration had discarded. Seven of 24 held-out angles sat where the model says brown, green or blue while the key said yellow, orange or purple. The model was right; the key was wrong. The defect was conservative, penalizing the word condition at least as much as the numeric one, and the registered analysis stands; the rule was still wrong and the amendment says so.

Two index spaces. The cross-model parser returned an index into the shuffled option list while the target indexed the canonical alphabet. Every cross-model word score sat at chance, four models at once, and it looked like four weak models. Only the word rung was affected, because the label rungs used slot indices on both sides, which is exactly why it looked like a property of the models. After the fix the two readouts agree on 97–100% of trials.

Output truncated before the answer. The first frontier run returned empty strings on ten of ten calls: the model spends its output budget on reasoning tokens before the answer, and the budget was eight tokens. A capable model would have been recorded as producing nothing. It was caught only because the raw text was printed beside the score.

None was found by looking at the score; all were found by looking at what the model said. The repository's audits now parse the prompt against the target, recompute every label from raw output, break the trial-to-answer pairing and require chance, and refuse alphabets the model does not emit.

9 LIMITATIONS

The real-image results come from one dataset, COCO, with categories skewed toward *person*, and one task: choosing among described grid cells for a named object. That is a selection task, a stand-in for how multiple-choice spatial benchmarks are written, not a grounding benchmark with free-form box output, and nothing here licenses a claim about box regression. A 3×3 and a 4×4 grid are coarse. The color results use synthetic squares on a gray field. Two frontier models were tested and only one shows the effect; a third might go either way. InternVL3-2B shares a Qwen language backbone, so the cross-family replication is partial. The frontier models are scored by generating and parsing while the local models are scored by first token; the two agree wherever both are available, but they are not the same instrument. The primary model runs at 4-bit precision, and although 8-bit widens the gap, full precision was not run. All decoding is greedy, which is standard for evaluation and is what the benchmarks we compare against use, but sampling was not tested. The prompts state each convention once, in one wording; whether a longer instruction that spells out the mapping narrows the gap was not tested. And readership, the quantity we introduce, is a valid test of whether a convention is read and a failed predictor of accuracy; the pre-registered out-of-sample test missed on every part.

10 WHAT A BENCHMARK AUTHOR SHOULD DO

Report the convention with the number; a localization score without the coordinate system it was scored in is not interpretable across papers. Run the swap: describe the same alternatives a second way, in plain language, and report both, because a difference of more than a few points is a property of the instrument and belongs next to the result. Run the mispaired test before trusting a coordinate

readout; a model that follows the coordinate at chance or below against a wrong name is not reading it. Do not gloss coordinates to help, since that turns the task into a naming task and the number stops being about coordinates. Separate parse failures from wrong answers, because a score that counts both as wrong hides which is happening. Calibrate the answer vocabulary per model; a fixed word list is a choice about which model the benchmark is easiest for. And check the scorer against the raw output: every error in §8 was invisible from the score and obvious from the text.

ETHICS STATEMENT

The work uses a public dataset (COCO) and synthetic stimuli, involves no human subjects or personal data, and its intended effect is more accurate evaluation of publicly available models.

REPRODUCIBILITY STATEMENT

Every number and figure in this paper is produced by scripts from the raw per-trial logs, none entered by hand, and a check script confirms that every decimal in this manuscript appears in their output. Two audit scripts verify, among other things, that the option at each stored target index actually names the target when the prompt string itself is parsed, that every stored label recomputes from the raw model output, that breaking the trial-to-answer pairing drives accuracy to chance, and that no alphabet is scored that the model does not emit. All hypotheses, endpoints, exclusions, failure conditions and amendments are recorded with timestamps in a versioned analysis plan, including the two predictions that failed. Code, logs, prompts, the analysis plan and both audits are available from the author on request and will be released publicly.

USE OF AI ASSISTANCE

The author conceived the project and its experimental program, set the research question, and decided the direction at every stage: what to test, what to read, which framings to abandon when their evidence did not hold, which intermediate results to reject as insufficient, which outside critiques to act on, and what to claim and submit. The author specified each experiment and each analysis, directed the literature search and chose what to read from it, and required that every claim survive a plain-language explanation and an independent audit before it was accepted. The labor-intensive parts of the work, principally writing code to the author's specification, running batches of trials, retrieving candidate references, and producing first drafts of text and tables, were delegated to AI tools, mainly Claude via Claude Code, with ChatGPT and Perplexity used to obtain independent critiques of the work in progress. The author implemented the experimental program on personal hardware and accounts, reviewed the output of every delegated step, and revised the text throughout; the final wording, structure and claims are the author's. Five scoring errors described in Section 8 arose during AI-assisted implementation and were caught by the audits the author required. Every number in the paper is regenerated from raw logs by scripts in the repository, and the pre-registered hypotheses, including the two that failed, are recorded with timestamps in the same repository.

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A Additional observations

Readership across models. On COCO pixels, readership is $1.12\times$ for InternVL3-2B, $1.44\times$ for Qwen3-VL-4B, $1.89\times$ for GPT-4o, $4.86\times$ for Qwen2.5-VL-7B and $6.62\times$ for Gemini. Normalizing each model's convention gap by its headroom (cell-name accuracy minus chance), the two models that barely read pixels lose 85–93% of that headroom to the convention and the three that read them lose 16–19%. With five points this is an observation, not a fit, and it was not pre-registered.

The number below chance. Qwen2.5-VL-7B scores 5.0% on the normalized convention, below chance and below the 10.9% its own answer bias would produce against the target distribution. We first tested whether it applies a systematic transform to the grid: the identity, transpose, anti-transpose, both flips, three rotations, and row and column shifts. None explains it; the anti-transpose reaches 19% ($p = 7 \times 10^{-4}$) and accounts for under a fifth of responses. The eleventh hypothesis, prompted by an outside review after those ten had failed, is that the model reads the 0–999 numbers in its training units, as pixels. For each trial we take the option whose numbers, read as pixels, lie nearest the object's box center and ask whether the model chose it. It did on 53.0% of trials, against a null of 11.8% from the product of the model's marginal answer distribution and the predicted-slot distribution ($p = 7.8 \times 10^{-48}$, exact binomial against $1/9$). That option is the true target on only 2.5% of trials, which is why the score falls below chance rather than to it. The same test applied to Qwen3-VL-4B on pixel numbers, read as thousandths of the image, gives 28.0% against 11.4% ($p = 4.6 \times 10^{-11}$); applied to InternVL3-2B it gives 10.5% against 10.2% ($p = 0.64$). An accurate model can inflate this rate through trials where the misread option happens to be the target; excluding those leaves 54.4% for Qwen2.5-VL and 26.4% for Qwen3-VL. Applied to Gemini and GPT-4o in both directions, the test is null (0.0 to 15.0% against nulls of 10.9 to 11.5%, smallest $p = 0.07$): the misreading accounts for the open-model inversions and not for the frontier gaps. The script and its output are in the repository, and the protocol file records the test as post hoc.

B Prompt templates, option order, gate sensitivity, and instrument agreement

Real photographs. Every prompt is one turn of text under the image. The examples below are the verbatim prompts for one image (COCO 139, 640×426), taken from the logs. The nine cells are listed in an order fixed per image and identical across conventions.

cell name, locate. `In this image, the microwave is at one of these positions: 1=center, 2=bottom-right, 3=middle-left, 4=top-right, 5=top-center, 6=middle-right, 7=bottom-center, 8=top-left, 9=bottom-left. Which one? Answer with a single integer 1 to 9 and nothing else.`

absolute pixels, locate. `Positions are given as (x, y) pixel coordinates in this 640 by 426 image, where (0, 0) is the top-left corner and x increases to the right and y increases downward. In this image, the microwave is at one of these positions: 1=(320, 213), 2=(533, 355), 3=(106, 213), 4=(533, 71), 5=(320, 71), 6=(533, 213), 7=(320, 355), 8=(106, 71), 9=(106, 355). Which one? Answer with a single integer 1 to 9 and nothing else.`

normalized, locate. `Positions are given as (x, y) coordinates normalized to a 0 to 999 range, where (0, 0) is the top-left corner and x increases to the right and y increases downward. In this image, the microwave is at one of these positions: 1=(500, 500), 2=(833, 833), 3=(166, 500), 4=(833, 167), 5=(500, 167), 6=(833, 500), 7=(500, 833), 8=(166, 167), 9=(166, 833). Which one? Answer with a single integer 1 to 9 and nothing else.`

glossed pixels, locate. `[same preamble] ... 1=(320, 213) [center], 2=(533, 355) [bottom-right], 3=(106, 213) [middle-left], ...`

mispaired pixels, locate. `[same preamble] ... 1=(320, 213) [middle-left], 2=(533, 355) [top-right], 3=(106, 213) [top-left], ...`

cell name, identify (coordinate on the question side). `In this image, what object is at position middle-right? It is one of these: 1=fork, 2=kite, 3=bottle, 4=microwave, 5=baseball glove, 6=donut. Answer with a single integer 1 to 6 and nothing else.`

absolute pixels, identify. `[same preamble] In this image, what object is at position (533, 213)? It is one of these: 1=fork, 2=kite, 3=bottle, 4=microwave, 5=baseball glove, 6=donut. Answer with a single integer 1 to 6 and nothing else.`

Color ladder. Templates, with the slots the harness fills in braces. {where} is a position word such as bottom-center; {options} is the K calibrated words in the trial's order; {which} is "Which of these is it closest to?" on held-out angles; {angle} is the model's own calibrated angle for each word (Table 5).

word. `In the array you memorized, the square at the {where} of the image was one of these colors: {options}. {which} Answer with a single color word and nothing else.`

label. `... one of these colors: 1={option 1}, 2={option 2}, ... {which} Answer with a single integer 1 to {K} and nothing else.`

numeric. `Colors are described by a hue angle from 0 to 359 on a color wheel where {angle} is {word}, ... In the array you memorized, the square at the {where} of the image had one of these hues: 1={angle 1}, 2={angle 2}, ... {which} Answer with a single integer 1 to {K} and nothing else.`

numeric, bare. `As numeric without the legend sentence.`

numeric, glossed. `... had one of these hues: 1={angle 1} ({word 1}), 2={angle 2} ({word 2}), ...`

numeric, mispaired. `As glossed, with the words permuted so that no angle carries its own word.`

Option order. For each trial the order of the alternatives is a deterministic permutation seeded by a hash of the trial identifier, so it is random across trials and identical across conventions for the same trial. The mispaired gloss is a derangement of the same seed, a permutation with no fixed point, so no option is ever correctly glossed by chance. Orders were not re-sampled; the paired design removes order effects from every within-item contrast, and the selection-bias literature (§2) is the place to look for their marginal size.

Gate sensitivity. The gate admits a model whose control condition is at least 20 points over chance. On the color task the models showing the penalty clear it by 38.2 to 67.7 points, so the set of penalty models is the same for every threshold from 0 to 38.2; only the denominator moves. LLaVA (null contrast) leaves above 20.8 and SmolVLM (null contrast) enters below 12.9. On the photographs the penalty models clear by 34.9 to 83.4 and the set is unchanged for any threshold up to 34.9; GPT-4o (null) leaves above 27.9. No conclusion in the paper depends on the value 20.

Instrument agreement. For every local model the crossmodel logs hold both readouts on every visible trial. First-token argmax and the parsed generation agree on 100.0% of trials for Qwen2.5-VL-7B and InternVL3-2B on all three rungs ($n = 71$ each), on 100.0% for LLaVA on the word rung ($n = 48$), and on 97.2% for SmolVLM on the word rung ($n = 71$; two trials). SmolVLM fails the gate and contributes no contrast.

C ADDITIONAL TABLES

Table 5: Calibrated color vocabularies. Anchors are wheel angles at the center of a run the model named unanimously across three renders.

model	stable anchors (angle, name)
Qwen3-VL-4B	5 pink, 50 orange, 95 yellow, 145 green, 245 blue, 310 purple
Qwen2.5-VL-7B	55 orange, 100 yellow, 140 green, 185 teal, 245 blue, 340 pink
InternVL3-2B	30 pink, 100 yellow, 145 green, 205 cyan, 255 blue, 310 purple
SmolVLM-2B	30 red, 105 yellow, 145 green, 205 cyan, 260 blue, 315 purple
LLaVA-1.5-7B	40 pink, 145 green, 235 blue, 310 purple

Table 6: COCO LOCATE, chance 11.1%, same 200 photographs. Gap is cell name minus pixel, paired per image. Gemini's gap is -15.5 counting unparseable answers as wrong and -11.1 on parseable answers only ($p = 1.0 \times 10^{-4}$); the pixel prompt produces 5.5% unparseable output against 0.0% for cell names. Frontier models as served on 20 September 2026.

model	cell name	normalized	pixel	gap	p
Qwen3-VL-4B	68.5	59.5	20.0	-48.5	1.2×10^{-21}
Qwen3-VL-4B, 8-bit	76.0	68.5	24.0	-52.0	5.0×10^{-24}
InternVL3-2B	46.0	19.0	13.5	-32.5	4.2×10^{-13}
Qwen2.5-VL-7B	68.0	5.0	59.0	-9.0	0.08
Gemini (API)	94.5	94.0	79.0	-15.5	3.4×10^{-7}
GPT-4o (API)	39.0	22.0	34.0	-5.0	0.24

Table 7: Robustness of the Qwen3-VL-4B cell-name minus pixel gap on COCO LOCATE, by tercile of object size (fraction of image area) and of the target center's distance from the nearest cell boundary (in cell widths). Exact McNemar, n images per slice.

slice		cell name	pixel	gap	p
object size	small (< 0.026)	66.7	18.2	-48.5	4.4×10^{-7}
	medium	77.3	24.2	-53.0	1.0×10^{-8}
	large (> 0.113)	61.8	17.6	-44.1	6.9×10^{-8}
boundary distance	near edge (< 0.30)	62.1	25.8	-36.4	7.0×10^{-5}
	middling	72.7	13.6	-59.1	2.2×10^{-10}
	well inside (> 0.36)	70.6	20.6	-50.0	5.4×10^{-9}
category	excluding *person*	—	—	-47.1	3.3×10^{-18}

Table 8: Synthetic spatial arrays, Qwen3-VL-4B, 120 trials, four colored squares in distinct cells of a 3×3 grid. LOCATE: 9 options, chance 11.1%; IDENTIFY: 6 options, chance 16.7%. Visible / cached. p is exact McNemar against cell names.

direction	options described as	visible	cached	p (visible / cached)
LOCATE	cell name	70.8	68.3	—
	normalized 0–999	58.3	48.3	0.05 / 1.2×10^{-3}
	absolute pixels	42.5	35.8	8.2×10^{-6} / 1.5×10^{-7}
IDENTIFY	cell name	96.7	95.8	—
	normalized 0–999	90.8	90.0	0.12 / 0.09
	absolute pixels	76.7	77.5	8.4×10^{-6} / 2.7×10^{-5}